

From: Jeff Dietlin <utilities@Cadillac-MI.net>
Sent: 1/23/2019 4:08:30 PM
To: "Berdinski, Thomas (DEQ)" <BERDINSKIT@michigan.gov>
Subject: FW: TestAmerica invoice and report files from 190-18507-1 WWTP - Biosolids
Attachments: J18507-1 UDS Level 2 Report Final Report.pdf

Tom,

Do you want the biosolids results for PFAS?

They are low.

Jeffrey M. Dietlin

City of Cadillac
Director of Utilities
(231) 779-7346(Direct)
(231)775-0906 (Fax)

From: Lab <lab@cadillac-mi.net>
Sent: Wednesday, January 23, 2019 3:54 PM
To: Doug Langworthy <dlangworthy@Cadillac-MI.net>; Jeff Dietlin <utilities@Cadillac-MI.net>
Subject: FW: TestAmerica invoice and report files from 190-18507-1 WWTP - Biosolids

Gentlemen –

Biosolids sample (PFAS) collected 12/27/18. The information which “matters directly to you” is on page 3-7 (of 16); the rest is mostly QA.

By my calculations, the PFOA and PFOS are in the 4 – 10 ppb range, on a dry weight basis.

If you have any questions, just let me know. Thanks,

Cindy A. Tomaszewski

Lab Supervisor

From: Schafer, Sue [<mailto:sue.schafer@testamericainc.com>]
Sent: Monday, January 21, 2019 5:11 PM
To: Lab
Subject: TestAmerica invoice and report files from 190-18507-1 WWTP - Biosolids

Hello,

Attached please find the invoice and report files for job 190-18507-1; WWTP - Biosolids

Please feel free to contact me if you have any questions.

Thank you.

Please let us know if we met your expectations by rating the service you received from TestAmerica on this project by visiting our website at: [Project Feedback](#)

SUE SCHAFFER
Project Manager

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

Tel: 810.229.2763
www.testamericainc.com

Reference: [042774]
Attachments: 2

ATTACHMENT NAME:

J18507-1 UDS Level 2 Report Final Report.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

ANALYTICAL REPORT

TestAmerica Laboratories, Inc.

TestAmerica Michigan

10448 Citation Drive

Suite 200

Brighton, MI 48116

Tel: (810)229-2763

TestAmerica Job ID: 190-18507-1

Client Project/Site: WWTP - Biosolids

For:

City of Cadillac Utilities

1121 Plett Road

Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:

1/21/2019 4:57:05 PM

Sue Schafer, Project Manager II

(810)229-2763

sue.schafer@testamericainc.com

LINKS

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results through

TotalAccess

Have a Question?



Visit us at:

www.testamericainc.com

This report has been electronically signed and authorized by the signatory. Electronic signature is intended to be the legally binding equivalent of a traditionally handwritten signature.

Results relate only to the items tested and the sample(s) as received by the laboratory.



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Sample Summary

Client: City of Cadillac Utilities
Project/Site: WWTP - Biosolids

TestAmerica Job ID: 190-18507-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-18507-1	Digester #4	Water	12/26/18 13:35	12/27/18 13:10
190-18507-2	Digester #4 Duplicate	Water	12/26/18 13:35	12/27/18 13:10
190-18507-3	Equipment Blank	Water	12/26/18 13:30	12/27/18 13:10
190-18507-4	Field Blank	Water	12/26/18 00:00	12/27/18 13:10

Case Narrative

Client: City of Cadillac Utilities
Project/Site: WWTP - Biosolids

TestAmerica Job ID: 190-18507-1

Job ID: 190-18507-1

Laboratory: TestAmerica Michigan

Narrative

Job Narrative 190-18507-1

Comments

No additional comments.

Receipt

The samples were received on 12/27/2018 1:10 PM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 1.3° C.

No analytical or quality issues were noted, other than those described in the Definitions/Glossary page.

10448 Citation Drive Suite 200
Brighton, MI 48116
Phone (810) 229-2763 Fax (810) 229-0000

2022376

Chain of Custody Record



TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

[illegible]

Sample Administration
Receipt Documentation Log

Doc Log ID: 237317



Group Number(s): 2022376

Client: Test America

Delivery and Receipt Information

Delivery Method:	<u>Fed Ex</u>	Arrival Timestamp:	<u>12/28/2018 10:00</u>
Number of Packages:	<u>1</u>	Number of Projects:	<u>1</u>

Arrival Condition Summary

Shipping Container Sealed:	Yes	Sample IDs on COC match Containers:	Yes
Custody Seal Present:	Yes	Sample Date/Times match COC:	Yes
Custody Seal Intact:	Yes	VOA Vial Headspace $\geq$ 6mm:	N/A
Samples Chilled:	Yes	Total Trip Blank Qty:	0
Paperwork Enclosed:	Yes	Air Quality Samples Present:	No
Samples Intact:	Yes		
Missing Samples:	No		
Extra Samples:	No		
Discrepancy in Container Qty on COC:	No		

Unpacked by Nicole Reiff (25684) at 13:52 on 12/28/2018

Samples Chilled Details

Thermometer Types: DT = Digital (Temp. Bottle) IR = Infrared (Surface Temp) All Temperatures in °C.

<u>Cooler #</u>	<u>Thermometer ID</u>	<u>Corrected Temp</u>	<u>Therm. Type</u>	<u>Ice Type</u>	<u>Ice Present?</u>	<u>Ice Container</u>	<u>Elevated Temp?</u>
1	32170023	0.6	IR	Wet	Y	Loose	N



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ANALYSIS REPORT

Prepared by:

Eurofins Lancaster Laboratories Environmental
2425 New Holland Pike
Lancaster, PA 17601

Prepared for:

TestAmerica Brighton MI
10448 Citation Drive
Suite 200
Brighton MI 48116

Report Date: January 18, 2019 11:34

Project: WWTP - Biosolids

Account #: 01042
Group Number: 2022376
PO Number: 19001884
Release Number: 190-18507-1
State of Sample Origin: MI

Electronic Copy To TestAmerica

Attn: Sue Schafer

Respectfully Submitted,



Wendy A. Kozma
Principal Specialist Group Leader

(717) 556-7257

To view our laboratory's current scopes of accreditation please go to <https://www.eurofinsus.com/environment-testing/laboratories/eurofins-lancaster-laboratories-environmental/certifications-and-accreditations-eurofins-lancaster-laboratories-environmental/> . Historical copies may be requested through your project manager.



SAMPLE INFORMATION

<u>Client Sample Description</u>	<u>Sample Collection Date/Time</u>	<u>ELLE#</u>
Digester #4 (190-18507-1) Water	12/26/2018 13:35	9956976
Digester #4 Duplicate (190-18507-2) Water	12/26/2018 13:35	9956977
Equipment Blank (190-18507-3) Water	12/26/2018 13:30	9956978
Field Blank (190-18507-4) Water	12/26/2018	9956979

The specific methodologies used in obtaining the enclosed analytical results are indicated on the Laboratory Sample Analysis Record.

Analysis Report

Sample Description: Digester #4 (190-18507-1) Water
WWTP - Biosolids

TestAmerica Brighton MI
ELLE Sample #: WW 9956976
ELLE Group #: 2022376
Matrix: Water

Project Name: WWTP - Biosolids

Submittal Date/Time: 12/28/2018 10:00

Collection Date/Time: 12/26/2018 13:35

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	200	600	1
14473	6:2 fluorotelomersulfonate	27619-97-2	N.D.	200	400	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	400	1,200	1
14473	NEtFOSAA	2991-50-6	N.D.	200	600	1
NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.						
14473	NMeFOSAA	2355-31-9	470 J	200	600	1
NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.						
14473	Perfluorononanesulfonate(Pfns)	68259-12-1	N.D.	120	400	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	400	1,200	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	82 J	60	200	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	180	400	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	100	400	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	120	400	1
14473	Pfhea-Perfluoroheptanoic Acid	375-85-9	N.D.	80	200	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	80	400	1
14473	Pfha-Perfluorohexanoic Acid	307-24-4	400 J	80	400	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	80	400	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	80	400	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	130 J	60	200	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	100	600	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	190 J	80	400	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	400	1,200	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	80	400	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	60	200	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	80	200	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	80	400	1

The recovery for a target analyte(s) in the Laboratory Control Spike(s) is outside the QC acceptance limits as noted on the QC Summary. Since the recovery is high and the target analyte(s) was not detected in the sample, the data is reported.

The sample injection internal standard peak areas were outside of the QC limits for both the initial injection and the re-injection. The values here are from the initial injection of the sample.

Reporting limits were raised due to interference from the sample matrix.

Wet Chemistry		SM 2540 G-1997	%	%	%	
		%Moisture Calc				
00111	Moisture	n.a.	97.3	0.50	0.50	1
Moisture represents the loss in weight of the sample after oven drying at 103 - 105 degrees Celsius. The moisture result reported is on an as-received basis.						

*=This limit was used in the evaluation of the final result

Sample Description: Digester #4 (190-18507-1) Water
WWTP - Biosolids

TestAmerica Brighton MI
ELLE Sample #: WW 9956976
ELLE Group #: 2022376
Matrix: Water

Project Name: WWTP - Biosolids

Submittal Date/Time: 12/28/2018 10:00

Collection Date/Time: 12/26/2018 13:35

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	19014016	01/16/2019 10:50	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	19014016	01/14/2019 12:15	Robert Brown	1
00111	Moisture	SM 2540 G-1997 %Moisture Calc	1	19003820005A	01/03/2019 13:04	Larry E Bevins	1

*=This limit was used in the evaluation of the final result

Analysis Report

Sample Description: Digester #4 Duplicate (190-18507-2) Water
WWTP - Biosolids

TestAmerica Brighton MI
ELLE Sample #: WW 9956977
ELLE Group #: 2022376
Matrix: Water

Project Name: WWTP - Biosolids

Submittal Date/Time: 12/28/2018 10:00

Collection Date/Time: 12/26/2018 13:35

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	200	600	1
14473	6:2 fluorotelomersulfonate	27619-97-2	N.D.	200	400	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	400	1,200	1
14473	NEtFOSAA	2991-50-6	210 J	200	600	1
NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.						
14473	NMeFOSAA	2355-31-9	590 J	200	600	1
NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.						
14473	Perfluorononanesulfonate (PfnS)	68259-12-1	N.D.	120	400	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	400	1,200	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	95 J	60	200	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	180	400	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	100	400	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	120	400	1
14473	Pfhea-Perfluoroheptanoic Acid	375-85-9	N.D.	80	200	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	80	400	1
14473	Pfha-Perfluoroheptanoic Acid	307-24-4	420	80	400	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	80	400	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	80	400	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	140 J	60	200	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	100	600	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	220 J	80	400	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	400	1,200	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	80	400	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	60	200	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	80	200	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	80	400	1

The recovery for a target analyte(s) in the Laboratory Control Spike(s) is outside the QC acceptance limits as noted on the QC Summary. Since the recovery is high and the target analyte(s) was not detected in the sample, the data is reported.

Reporting limits were raised due to interference from the sample matrix.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	19014016	01/16/2019 10:59	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	19014016	01/14/2019 12:15	Robert Brown	1

*=This limit was used in the evaluation of the final result

Analysis Report

Sample Description: Equipment Blank (190-18507-3) Water
WWTP - Biosolids

TestAmerica Brighton MI
ELLE Sample #: WW 9956978
ELLE Group #: 2022376
Matrix: Water

Project Name: WWTP - Biosolids

Submittal Date/Time: 12/28/2018 10:00

Collection Date/Time: 12/26/2018 13:30

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	0.94	2.8	1
14473	6:2 fluorotelomersulfonate	27619-97-2	N.D.	0.94	1.9	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	1.9	5.7	1
14473	NEtFOSAA	2991-50-6	N.D.	0.94	2.8	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	N.D.	0.94	2.8	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorononanesulfonate(Pfns)	68259-12-1	N.D.	0.57	1.9	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.9	5.7	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	N.D.	0.28	0.94	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.85	1.9	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.47	1.9	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	0.57	1.9	1
14473	Pfhp-Perfluoroheptanoic Acid	375-85-9	N.D.	0.38	0.94	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.38	1.9	1
14473	Pfha-Perfluoroheptanoic Acid	307-24-4	N.D.	0.38	1.9	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	0.38	1.9	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.38	1.9	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	0.32 J	0.28	0.94	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.47	2.8	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	N.D.	0.38	1.9	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.9	5.7	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.38	1.9	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.28	0.94	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.38	0.94	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.38	1.9	1

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	19003001	01/04/2019 17:24	Christine E Dolman	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	19003001	01/03/2019 07:55	Courtney J Fatta	1

*=This limit was used in the evaluation of the final result

Analysis Report

Sample Description: Field Blank (190-18507-4) Water
WWTP - Biosolids

TestAmerica Brighton MI
ELLE Sample #: WW 9956979
ELLE Group #: 2022376
Matrix: Water

Project Name: WWTP - Biosolids

Submittal Date/Time: 12/28/2018 10:00

Collection Date/Time: 12/26/2018

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit*	Limit of Quantitation	Dilution Factor
LC/MS/MS Miscellaneous		EPA 537 Version 1.1 Modified	ng/l	ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	0.84	2.5	1
14473	6:2 fluorotelomersulfonate	27619-97-2	N.D.	0.84	1.7	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	1.7	5.1	1
14473	NEtFOSAA	2991-50-6	N.D.	0.84	2.5	1
NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.						
14473	NMeFOSAA	2355-31-9	N.D.	0.84	2.5	1
NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.						
14473	Perfluorononanesulfonate(Pfns)	68259-12-1	N.D.	0.51	1.7	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.7	5.1	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	N.D.	0.25	0.84	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.76	1.7	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.42	1.7	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	0.51	1.7	1
14473	Pfpha-Perfluoroheptanoic Acid	375-85-9	N.D.	0.34	0.84	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.34	1.7	1
14473	Pfha-Perfluoroheptanoic Acid	307-24-4	N.D.	0.34	1.7	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	0.34	1.7	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.34	1.7	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	N.D.	0.25	0.84	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.42	2.5	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	N.D.	0.34	1.7	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.7	5.1	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.34	1.7	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.25	0.84	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.34	0.84	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.34	1.7	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits as noted on the QC Summary.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	19003001	01/04/2019 17:34	Christine E Dolman	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	19003001	01/03/2019 07:55	Courtney J Fatta	1

*=This limit was used in the evaluation of the final result

Quality Control Summary

Client Name: TestAmerica Brighton MI
Reported: 01/18/2019 11:34

Group Number: 2022376

Matrix QC may not be reported if insufficient sample or site-specific QC samples were not submitted. In these situations, to demonstrate precision and accuracy at a batch level, a LCS/LCSD was performed, unless otherwise specified in the method.

All Inorganic Initial Calibration and Continuing Calibration Blanks met acceptable method criteria unless otherwise noted on the Analysis Report.

Method Blank

Analysis Name	Result	MDL**	LOQ
	ng/l	ng/l	ng/l
Batch number: 19003001	Sample number(s): 9956978-9956979		
4:2 fluorotelomersulfonate	N.D.	1.0	3.0
6:2 fluorotelomersulfonate	N.D.	1.0	2.0
8:2 fluorotelomersulfonate	N.D.	2.0	6.0
NEtFOSAA	N.D.	1.0	3.0
NMeFOSAA	N.D.	1.0	3.0
Perfluorononanesulfonate(Pfns)	N.D.	0.60	2.0
Pfba-Perfluorobutanoic Acid	N.D.	2.0	6.0
Pfbs-Perfluorobutanesulfonate	N.D.	0.30	1.0
Pfda-Perfluorodecanoic Acid	N.D.	0.90	2.0
Pfdoda-Perfluorododecanoic	N.D.	0.50	2.0
Pfds-Perfluorodecanesulfonate	N.D.	0.60	2.0
Pfhpa-Perfluoroheptanoic Acid	N.D.	0.40	1.0
Pfhps-Perfluoroheptanesulfonat	N.D.	0.40	2.0
Pfhxa-Perfluorohexanoic Acid	N.D.	0.40	2.0
Pfhxs-Perfluorohexanesulfonate	N.D.	0.40	2.0
Pfna-Perfluorononanoic Acid	N.D.	0.40	2.0
Pfoa-Perfluorooctanoic Acid	N.D.	0.30	1.0
Pfosa-Perfluorooctanesulfonami	N.D.	0.50	3.0
Pfos-Perfluorooctanesulfonate	N.D.	0.40	2.0
Pfpea-Perfluoropentanoic Acid	N.D.	2.0	6.0
Pfpes-Perfluoropentanesulfonat	N.D.	0.40	2.0
Pfteda-Perfluorotetradecanoic	N.D.	0.30	1.0
Pftrda-Perfluorotridecanoic Ac	N.D.	0.40	1.0
Pfunda-Perfluoroundecanoic Aci	N.D.	0.40	2.0
Batch number: 19014016	Sample number(s): 9956976-9956977		
4:2 fluorotelomersulfonate	N.D.	200	600
6:2 fluorotelomersulfonate	N.D.	200	400
8:2 fluorotelomersulfonate	N.D.	400	1,200
NEtFOSAA	N.D.	200	600
NMeFOSAA	N.D.	200	600
Perfluorononanesulfonate(Pfns)	N.D.	120	400
Pfba-Perfluorobutanoic Acid	N.D.	400	1,200
Pfbs-Perfluorobutanesulfonate	N.D.	60	200
Pfda-Perfluorodecanoic Acid	N.D.	180	400
Pfdoda-Perfluorododecanoic	N.D.	100	400
Pfds-Perfluorodecanesulfonate	N.D.	120	400
Pfhpa-Perfluoroheptanoic Acid	N.D.	80	200
Pfhps-Perfluoroheptanesulfonat	N.D.	80	400

*- Outside of specification

**--This limit was used in the evaluation of the final result for the blank

(1) The result for one or both determinations was less than five times the LOQ.

(2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica Brighton MI
Reported: 01/18/2019 11:34

Group Number: 2022376

Method Blank (continued)

Analysis Name	Result ng/l	MDL** ng/l	LOQ ng/l
Pfha-Perfluorohexanoic Acid	N.D.	80	400
Pfhxs-Perfluorohexanesulfonate	N.D.	80	400
Pfna-Perfluorononanoic Acid	N.D.	80	400
Pfoa-Perfluorooctanoic Acid	N.D.	60	200
Pfosa-Perfluorooctanesulfonami	N.D.	100	600
Pfos-Perfluorooctanesulfonate	N.D.	80	400
Pfpea-Perfluoropentanoic Acid	N.D.	400	1,200
Pfpes-Perfluoropentanesulfonat	N.D.	80	400
Pfteda-Perfluorotetradecanoic	N.D.	60	200
Pftrda-Perfluorotridecanoic Ac	N.D.	80	200
Pfunda-Perfluoroundecanoic Aci	N.D.	80	400

LCS/LCSD

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Batch number: 19003001	Sample number(s): 9956978-9956979								
4:2 fluorotelomersulfonate	14.94	17.76	14.94	15.18	119	102	82-152	16	30
6:2 fluorotelomersulfonate	15.17	16.21	15.17	17.12	107	113	66-155	5	30
8:2 fluorotelomersulfonate	15.33	16.02	15.33	13.62	104	89	66-148	16	30
NEtFOSAA	5.44	6.00	5.44	6.27	110	115	55-169	5	30
NMeFOSAA	5.44	5.74	5.44	6.27	105	115	44-147	9	30
Perfluorononanesulfonate(Pfns)	5.22	6.20	5.22	6.13	119	117	66-133	1	30
Pfba-Perfluorobutanoic Acid	5.44	6.89	5.44	6.26	127	115	74-142	10	30
Pfbs-Perfluorobutanesulfonate	4.81	5.75	4.81	5.32	120	111	73-128	8	30
Pfda-Perfluorodecanoic Acid	5.44	6.33	5.44	5.04	116	93	69-148	23	30
Pfdoda-Perfluorododecanoic	5.44	6.30	5.44	5.44	116	100	75-136	15	30
Pfds-Perfluorodecanesulfonate	5.24	5.88	5.24	5.67	112	108	60-135	4	30
Pfhpa-Perfluoroheptanoic Acid	5.44	6.81	5.44	6.82	125	125	76-140	0	30
Pfhps-Perfluoroheptanesulfonat	5.18	6.52	5.18	5.76	126	111	64-135	12	30
Pfhxa-Perfluorohexanoic Acid	5.44	6.77	5.44	6.10	124	112	75-135	10	30
Pfhxs-Perfluorohexanesulfonate	5.14	5.93	5.14	5.29	115	103	71-131	12	30
Pfna-Perfluorononanoic Acid	5.44	5.83	5.44	6.42	107	118	72-148	10	30
Pfoa-Perfluorooctanoic Acid	5.44	6.28	5.44	5.90	115	108	72-138	6	30
Pfosa-Perfluorooctanesulfonami	5.44	6.01	5.44	4.95	110	91	65-164	19	30
Pfos-Perfluorooctanesulfonate	5.20	5.06	5.20	5.07	97	97	67-138	0	30
Pfpea-Perfluoropentanoic Acid	5.44	6.10	5.44	6.32	112	116	74-134	4	30
Pfpes-Perfluoropentanesulfonat	5.10	6.74	5.10	5.61	132*	110	76-127	18	30
Pfteda-Perfluorotetradecanoic	5.44	5.90	5.44	5.63	108	103	74-135	5	30
Pftrda-Perfluorotridecanoic Ac	5.44	6.59	5.44	5.98	121	110	61-145	10	30
Pfunda-Perfluoroundecanoic Aci	5.44	7.14	5.44	5.46	131	100	75-146	27	30

*- Outside of specification

**--This limit was used in the evaluation of the final result for the blank

(1) The result for one or both determinations was less than five times the LOQ.

(2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica Brighton MI
Reported: 01/18/2019 11:34

Group Number: 2022376

LCS/LCSD (continued)

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Batch number: 19014016	Sample number(s): 9956976-9956977								
4:2 fluorotelomersulfonate	1494.4	1779.28	1494.4	1660.38	119	111	82-152	7	30
6:2 fluorotelomersulfonate	1516.8	1654.32	1516.8	1525.1	109	101	66-155	8	30
8:2 fluorotelomersulfonate	1532.8	1661.26	1532.8	1941.09	108	127	66-148	16	30
NEtFOSAA	544	478.37	544	577.7	88	106	55-169	19	30
NMeFOSAA	544	500.42	544	580.16	92	107	44-147	15	30
Perfluorononanesulfonate(Pfns)	522.4	634.77	522.4	573.31	122	110	66-133	10	30
Pfba-Perfluorobutanoic Acid	544	786.87	544	785.33	145*	144*	74-142	0	30
Pfbs-Perfluorobutanesulfonate	481.6	521.33	481.6	498.85	108	104	73-128	4	30
Pfda-Perfluorodecanoic Acid	544	633.19	544	593.24	116	109	69-148	7	30
Pfdoda-Perfluorododecanoic	544	641.21	544	564.75	118	104	75-136	13	30
Pfds-Perfluorodecanesulfonate	524	485.75	524	574.13	93	110	60-135	17	30
Pfhpa-Perfluoroheptanoic Acid	544	628.95	544	611.43	116	112	76-140	3	30
Pfhps-Perfluoroheptanesulfonat	517.6	581.27	517.6	584.4	112	113	64-135	1	30
Pfhxa-Perfluorohexanoic Acid	544	589.17	544	584.03	108	107	75-135	1	30
Pfhxs-Perfluorohexanesulfonate	514.4	528.78	514.4	509.81	103	99	71-131	4	30
Pfna-Perfluorononanoic Acid	544	571.02	544	574.63	105	106	72-148	1	30
Pfoa-Perfluorooctanoic Acid	544	550.71	544	562.08	101	103	72-138	2	30
Pfosa-Perfluorooctanesulfonami	544	533.21	544	496.87	98	91	65-164	7	30
Pfos-Perfluorooctanesulfonate	520	492.15	520	501.94	95	97	67-138	2	30
Pfpea-Perfluoropentanoic Acid	544	626.41	544	626.95	115	115	74-134	0	30
Pfpes-Perfluoropentanesulfonat	510.4	573.35	510.4	556.23	112	109	76-127	3	30
Pfteda-Perfluorotetradecanoic	544	617.03	544	594.46	113	109	74-135	4	30
Pftrda-Perfluorotridecanoic Ac	544	485.48	544	504.08	89	93	61-145	4	30
Pfunda-Perfluoroundecanoic Aci	544	566.28	544	553.8	104	102	75-146	2	30
	%	%	%	%					
Batch number: 19003820005A	Sample number(s): 9956976								
Moisture	89.5	89.43			100		99-101		

Labeled Isotope Quality Control

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 19003001

	13C4-PFBA	13C5-PFPeA	13C3-PFBS	13C2-4:2-FTS	13C5-PFHxA	13C3-PFHxS
9956978	82	80	78	140	76	85
9956979	91	91	86	89	85	95
Blank	79	82	71	76	75	90

*- Outside of specification

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(1) The result for one or both determinations was less than five times the LOQ.

(2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica Brighton MI
Reported: 01/18/2019 11:34

Group Number: 2022376

Labeled Isotope Quality Control

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 19003001

	13C4-PFBA	13C5-PFPeA	13C3-PFBS	13C2-4:2-FTS	13C5-PFHxA	13C3-PFHxS
LCS	67	66	60	73	64	59
LCSD	75	70	71	85	72	72
Limits:	33-123	31-157	26-148	21-182	35-138	34-126
	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
9956978	85	134	80	85	90	83
9956979	86	108	90	88	84	92
Blank	78	96	81	78	77	77
LCS	64	87	68	68	66	67
LCSD	72	92	77	75	77	83
Limits:	35-126	32-170	48-122	50-121	41-144	47-125
	13C2-8:2-FTS	d3-NMeFOSAA	13C7-PFUnDA	d5-NeIFOSAA	13C2-PFDoDA	13C2-PFTeDA
9956978	149	107	74	137	55	73
9956979	93	130*	86	132	85	84
Blank	79	101	79	119	72	77
LCS	81	59	63	71	59	64
LCSD	96	70	82	78	73	74
Limits:	27-164	30-127	30-128	30-142	39-130	26-119
	13C8-PFOA					
9956978	70					
9956979	77					
Blank	65					
LCS	63					
LCSD	73					
Limits:	11-127					

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 19014016

	13C4-PFBA	13C5-PFPeA	13C3-PFBS	13C2-4:2-FTS	13C5-PFHxA	13C3-PFHxS
9956976	91	86	73	84	75	92
9956977	97	96	83	85	75	94
Blank	97	97	91	94	102	100
LCS	93	93	90	87	95	95
LCSD	90	90	90	84	96	94

*- Outside of specification

**--This limit was used in the evaluation of the final result for the blank

(1) The result for one or both determinations was less than five times the LOQ.

(2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica Brighton MI
Reported: 01/18/2019 11:34

Group Number: 2022376

Labeled Isotope Quality Control (continued)

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 19014016

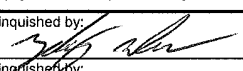
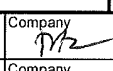
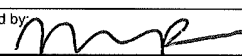
Limits:	33-123	31-157	26-148	21-182	35-138	34-126
	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
9956976	99	119	96	91	109	95
9956977	104	129	102	94	102	100
Blank	104	103	100	91	95	89
LCS	97	102	97	90	84	84
LCSD	96	106	98	95	90	89
Limits:	35-126	32-170	48-122	50-121	41-144	47-125
	13C2-8:2-FTS	d3-NMeFOSAA	13C7-PFUnDA	d5-NEtFOSAA	13C2-PFDoDA	13C2-PFTeDA
9956976	126	87	86	104	90	93
9956977	120	89	95	95	99	89
Blank	94	69	82	76	66	42
LCS	97	71	84	67	70	43
LCSD	85	71	86	74	76	50
Limits:	27-164	30-127	30-128	30-142	39-130	26-119
	13C8-PFOSA					
9956976	69					
9956977	88					
Blank	88					
LCS	86					
LCSD	90					
Limits:	11-127					

*- Outside of specification

**--This limit was used in the evaluation of the final result for the blank

(1) The result for one or both determinations was less than five times the LOQ.

(2) The unspiked result was more than four times the spike added.

Client Information (Sub Contract Lab)		Sample:		Lab PM: Schafer, Sue		Carrier Tracking No(s):		COC No: 190-21609.1																					
Client Contact: Shipping/Receiving		Phone:		E-Mail: sue.schafer@testamericainc.com		State of Origin: Michigan		Page: Page 1 of 1																					
Company: Eurofins Lancaster Laboratories Env LLC				Accreditations Required (See note):				Job #: 190-18507-1																					
Address: 2425 New Holland Pike,		Due Date Requested: 1/9/2019		Analysis Requested				Preservation Codes: A - HCL M - Hexane B - NaOH N - None C - Zn Acetate O - AsNaO2 D - Nitric Acid P - Na2O4S E - NaHSO4 Q - Na2SO3 F - MeOH R - Na2S2O3 G - Amchlor S - H2SO4 H - Ascorbic Acid T - TSP Dodecahydrate I - Ice U - Acetone J - DI Water V - MCAA K - EDTA W - pH 4-5 L - EDA Z - other (specify)																					
City: Lancaster		TAT Requested (days):																											
State, Zip: PA, 17601		PO #:		Field Filtered Sample (Yes or No) Perform MS/MSD (Yes or No) SUB (PFAS)/ PFAS				Total Number of containers																					
Phone: 717-656-2300(Tel)		WO #:																											
Email:		Project #:		Special Instructions/Note:																									
Project Name: WWTP - Biosolids		SSOW#:																											
Site:																													
Sample Identification - Client ID (Lab ID)		Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/soil, BT=Tissue, A=Air)																								
Digester #4 (190-18507-1)		12/26/18	13:35 Eastern		Water		X											3	METHOD 537M - 24 PFAS COMPOUNDS										
Digester #4 Duplicate (190-18507-2)		12/26/18	13:35 Eastern		Water		X											2	METHOD 537M - 24 PFAS COMPOUNDS										
Equipment Blank (190-18507-3)		12/26/18	13:30 Eastern		Water		X											2	METHOD 537M - 24 PFAS COMPOUNDS										
Field Blank (190-18507-4)		12/26/18	Eastern		Water		X											2	METHOD 537M - 24 PFAS COMPOUNDS										
Note: Since laboratory accreditations are subject to change, TestAmerica Laboratories, Inc. places the ownership of method, analyte & accreditation compliance upon out subcontract laboratories. This sample shipment is forwarded under chain-of-custody. If the laboratory does not currently maintain accreditation in the State of Origin listed above for analysis/tests/matrix being analyzed, the samples must be shipped back to the TestAmerica laboratory or other instructions will be provided. Any changes to accreditation status should be brought to TestAmerica Laboratories, Inc. attention immediately. If all requested accreditations are current to date, return the signed Chain of Custody attesting to said compliance to TestAmerica Laboratories, Inc.																													
Possible Hazard Identification										Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)																			
Unconfirmed										☐ Return To Client ☐ Disposal By Lab ☐ Archive For _____ Months																			
Deliverable Requested: I, II, III, IV, Other (specify)										Primary Deliverable Rank: 2																			
										Special Instructions/QC Requirements:																			
Empty Kit Relinquished by:					Date:					Time:					Method of Shipment:														
Relinquished by: 					Date/Time: 12/27/18 16:09					Company: 					Received by: _____					Date/Time: _____					Company: _____				
Relinquished by: _____					Date/Time: _____					Company: _____					Received by: _____					Date/Time: _____					Company: _____				
Relinquished by: _____					Date/Time: _____					Company: _____					Received by: 					Date/Time: 12/28/18 10:00					Company: EUE				
Custody Seals Intact: ☒ Yes ☐ No					Custody Seal No.:					Cooler Temperature(s) °C and Other Remarks: 0.6																			

Sample Administration
Receipt Documentation Log

Doc Log ID: 237317



Group Number(s): 2022376

Client: Test America**Delivery and Receipt Information**

Delivery Method:	<u>Fed Ex</u>	Arrival Timestamp:	<u>12/28/2018 10:00</u>
Number of Packages:	<u>1</u>	Number of Projects:	<u>1</u>

Arrival Condition Summary

Shipping Container Sealed:	Yes	Sample IDs on COC match Containers:	Yes
Custody Seal Present:	Yes	Sample Date/Times match COC:	Yes
Custody Seal Intact:	Yes	VOA Vial Headspace $\geq$ 6mm:	N/A
Samples Chilled:	Yes	Total Trip Blank Qty:	0
Paperwork Enclosed:	Yes	Air Quality Samples Present:	No
Samples Intact:	Yes		
Missing Samples:	No		
Extra Samples:	No		
Discrepancy in Container Qty on COC:	No		

*Unpacked by Nicole Reiff (25684) at 13:52 on 12/28/2018***Samples Chilled Details***Thermometer Types: DT = Digital (Temp. Bottle) IR = Infrared (Surface Temp) All Temperatures in °C.*

<u>Cooler #</u>	<u>Thermometer ID</u>	<u>Corrected Temp</u>	<u>Therm. Type</u>	<u>Ice Type</u>	<u>Ice Present?</u>	<u>Ice Container</u>	<u>Elevated Temp?</u>
1	32170023	0.6	IR	Wet	Y	Loose	N

Explanation of Symbols and Abbreviations

The following defines common symbols and abbreviations used in reporting technical data:

BMQL	Below Minimum Quantitation Level	mL	milliliter(s)
C	degrees Celsius	MPN	Most Probable Number
cfu	colony forming units	N.D.	non-detect
CP Units	cobalt-chloroplatinate units	ng	nanogram(s)
F	degrees Fahrenheit	NTU	nephelometric turbidity units
g	gram(s)	pg/L	picogram/liter
IU	International Units	RL	Reporting Limit
kg	kilogram(s)	TNTC	Too Numerous To Count
L	liter(s)	µg	microgram(s)
lb.	pound(s)	µL	microliter(s)
m3	cubic meter(s)	umhos/cm	micromhos/cm
meq	milliequivalents	MCL	Maximum Contamination Limit
mg	milligram(s)		
<	less than		
>	greater than		
ppm	parts per million - One ppm is equivalent to one milligram per kilogram (mg/kg) or one gram per million grams. For aqueous liquids, ppm is usually taken to be equivalent to milligrams per liter (mg/l), because one liter of water has a weight very close to a kilogram. For gases or vapors, one ppm is equivalent to one microliter per liter of gas.		
ppb	parts per billion		
Dry weight basis	Results printed under this heading have been adjusted for moisture content. This increases the analyte weight concentration to approximate the value present in a similar sample without moisture. All other results are reported on an as-received basis.		

Analytical test results meet all requirements of the associated regulatory program (i.e., NELAC (TNI), DoD, and ISO 17025) unless otherwise noted under the individual analysis.

Measurement uncertainty values, as applicable, are available upon request.

Tests results relate only to the sample tested. Clients should be aware that a critical step in a chemical or microbiological analysis is the collection of the sample. Unless the sample analyzed is truly representative of the bulk of material involved, the test results will be meaningless. If you have questions regarding the proper techniques of collecting samples, please contact us. We cannot be held responsible for sample integrity, however, unless sampling has been performed by a member of our staff.

This report shall not be reproduced except in full, without the written approval of the laboratory.

Times are local to the area of activity. Parameters listed in the 40 CFR Part 136 Table II as "analyze immediately" are not performed within 15 minutes.

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Data Qualifiers

Qualifier	Definition
C	Result confirmed by reanalysis
D1	Indicates for dual column analyses that the result is reported from column 1
D2	Indicates for dual column analyses that the result is reported from column 2
E	Concentration exceeds the calibration range
K1	Initial Calibration Blank is above the QC limit and the sample result is ND
K2	Continuing Calibration Blank is above the QC limit and the sample result is ND
K3	Initial Calibration Verification is above the QC limit and the sample result is ND
K4	Continuing Calibration Verification is above the QC limit and the sample result is ND
J (or G, I, X)	Estimated value $\geq$ the Method Detection Limit (MDL or DL) and $<$ the Limit of Quantitation (LOQ or RL)
P	Concentration difference between the primary and confirmation column $>40\%$. The lower result is reported.
P^	Concentration difference between the primary and confirmation column $>40\%$. The higher result is reported.
U	Analyte was not detected at the value indicated
V	Concentration difference between the primary and confirmation column $>100\%$. The reporting limit is raised due to this disparity and evident interference.
W	The dissolved oxygen uptake for the unseeded blank is greater than 0.20 mg/L.
Z	Laboratory Defined - see analysis report

Additional Organic and Inorganic CLP qualifiers may be used with Form 1 reports as defined by the CLP methods. Qualifiers specific to Dioxin/Furans and PCB Congeners are detailed on the individual Analysis Report.

Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: WWTP - Biosolids

TestAmerica Job ID: 190-18507-1

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
α	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
PQL	Practical Quantitation Limit
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: WWTP - Biosolids

TestAmerica Job ID: 190-18507-1

Laboratory: TestAmerica Michigan

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	EPA Region	Identification Number	Expiration Date
Michigan	State Program	5	57	05-05-20

Chain of Custody Record

Client Information Client Contact: Cindy Tomaszewski Company: City of Cadillac Utilities Address: 1121 Plett Road City: Cadillac State/Zip: MI, 49601 Phone: 2049-0419 Email: lab@cadillac-mi.net Project Name: City of Cadillac Site: WWTP-BIOSOLIDS		Sampler: Dave Skiff Lab PM: Schaefer, Sue E-Mail: sue.schafer@testamericainc.com Due Date Requested: TAT Requested (days): PO #: 2049-0419 WO #: 19001884 Project #: 19001884 SSOW#:		Carrier Tracking No(s): 190-22415-977.1 Page: Page 1 of 1 Job #:	
Analysis Requested Due Date Requested: TAT Requested (days): PO #: 2049-0419 WO #: 19001884 Project #: 19001884 SSOW#:		Preservation Codes: A - HCL B - NaOH C - Zn Acetate D - Nitric Acid E - NaHSO4 F - MeOH G - Amchlor H - Ascorbic Acid I - Ice J - DI Water K - EDTA L - EDA Other: M - Hexane N - None O - AsNaO2 P - Na2O4S Q - Na2SO3 R - Na2SO3 S - H2SO4 T - TSP Dodecahydrate U - Acetone V - MCAA W - pH 4-5 Z - other (specify)			
Sample Identification DIGESTER #4 DIGESTER #4 DUPLIC EQUIPMENT BLANK FIELD BLANK		Sample Date 12-26-18 12-26-18 12-26-18 12-26-18		Sample Time 1335 1335 1330 —	
Sample Type (C=Comp, G=grab) G G G —		Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air) Solid Solid Water Water		Preservation Code: N N N N	
Field Filtered Sample (Yes or No) N N N N		Perform MS/MSD (Yes or No) N N N N		PFC, IDA - PFA, Standard List (24 Analytes) N N N N	
Total Number of Containers 3 2 2 2		Special Instructions/Note: wastewater sludge u u		Special Instructions/Note: 190-18507 Chain of Custody	
Possible Hazard Identification ☐ Non-Hazard ☐ Flammable ☐ Skin Irritant ☐ Poison B ☒ Unknown ☐ Radiological Waste Deliverable Requested: I, II, III, IV, Other (specify)					
Empty Kit Relinquished by:					
Relinquished by: City of Cadillac Date/Time: 12-26-2018/1410 Relinquished by: UPS Date/Time: 12-27-18 13:10 Relinquished by:					
Custody Seals Intact: Δ Yes Δ No					
Custody Seal No.:					

TestAmerica

THE LEADER IN ENVIRONMENTAL TESTING

Cooler/Sample Receipt

(AFTER HOURS receipt: complete gray areas)
Place cooler in walk-in; place this form in Receiving in box. Date Time rec'd Initials

- ☐ MSDS or Known Hazard Information Supplied by Client
☐ Bottle stickers applied ☐ ELEMENT comment entered ☐ MSDS COC scanned emailed to EH&S
☐ Discrepancies Client ID City of Cadillac
☐ Short Hold Work Order # 190-18507
☐ Rush ☐ 24hr ☐ 2day ☐ 3day ☐ 5day ☐ Other
 Receipt evaluation performed by - Initials AMY Date 12/27/18 Time 13:10

Method of Shipment:

- ☐ Walk-In Client ☐ TestAmerica Field/Courier
☐ Other Client/3rd Party Courier
☐ Fed Ex Tracking #
☒ UPS Tracking # 1ZE47 491 03 5370 8647
☐ Other Ground

Shipping Container Type:

- ☒ Cooler ☐ Box
☐ None ☐ Other
 Packing Materials:
☒ Plastic Bags ☐ Foam
☐ Bubble Wrap ☐ Paper
☐ Packing Peanuts ☐ None
☒ Other Arr Pcs

Custody Seals Intact:

- ☒ Yes 58473637 ☐ No
☐ N/A (not used or required)
 Cooling Materials:
☒ Ice (solid) ☐ Ice (Melted)
☐ Blue Ice ☐ None
☐ Other

Biological: Temp (°C) Corrected
 Samples

Frozen
 yes no

Received within 2 hours
 yes no

Sample Flagged
 yes no

Receipt Temperatures

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Received on same day	Acceptable?	Cooler ID	Note Affected Samples if temperature not acceptable
140252483			☐	☐	☐ Yes ☐ No	☐ Yes ☐ No		
140252476			☐	☐	☐ Yes ☐ No	☐ Yes ☐ No		
CP313207	2.0	1.3	☐	☒	☐ Yes ☐ No	☒ Yes ☐ No		

* Receipt temperatures are considered acceptable if the samples are received on the same day they were collected & show signs that the cooling process has started. Temperature acceptance for most tests is ≤6.0°C, but not frozen. For additional information, please refer to SOP DT-SCA-004 Sample Receipt and Login, Attachment 2 - Holding Times, Preservation and Container Requirements

Receipt Questions**

	Y	N	n/a	"No" answers require additional comment
COC present & TA receipt signature, date, & time properly documented?	☒			
Containers & labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	☒			
Appropriate containers used & adequate volume provided?	☒			
Number of sample containers match COC?	☒			
Samples received within hold time?	☒			
Samples submitted for GRO and Volatiles analyses (8260, 624, 524) received without headspace?			☒	
Was a Trip Blank received with VOA samples?			☒	
Were the samples free of any questionable physical conformities? For example, field duplicates or multiple bottles of the same sample do not significantly vary in appearance (color, proportion of solids, etc.)	☒			
Were the COC, bottle labels, and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	☒			

** May not be applicable if samples are not for compliance testing

* Excludes FOG, Volatiles, TOC Vials

Client Contact Record

Contact via: ☐ Phone ☐ Email ☐ Other Person Contacted: Date/Time:

☐ Discrepancy allowance agreement is on record in the client project file

Discussion/Resolution: per G. Wood email chain: analyze as a water; run TS on 1 sample only. Do not DW correct.

Any additional documentation and clarification from client must be noted in the narrative and/or scanned into the COC directory.

Reviewed by PM Signature

Date

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 effective 06/11/12